

Analyzer-angle signatures of linear and circular polarization

A synthetic teaching paper for testing Research Starter Paper Guide

Abstract

We compare idealized analyzer-angle scans for nominally linear and circular polarization. The linear series follows Malus' law, while the circular series remains nearly angle independent. A small offset and deterministic perturbation are included to illustrate baseline subtraction, uncertainty, model checking, and the distinction between an observation and a physical conclusion.

Questions addressed

1. Which angular signature distinguishes the two prepared states?
2. How does a detector offset alter normalized contrast?
3. Which conclusions require calibration evidence beyond the plotted curves?

Key equations

Linear model: $I(\theta) = I_0 \cos^2(\theta - \theta_0) + b$

Visibility: $V = (I_{\max} - I_{\min}) / (I_{\max} + I_{\min})$

1. Synthetic method

A source is prepared in two nominal states and measured after a rotating linear analyzer. Angles span 0 to 180 degrees in 15 degree steps. Each point represents the mean of five synthetic repeats. The detector offset b is estimated from a blocked-source reading and subtracted before normalization.

Parameter	Linear run	Circular run
Analyzer step	15 deg	15 deg
Synthetic repeats	5	5
Blocked offset	0.030	0.030
Mean normalized intensity	0.515	0.502
Visibility	0.91	0.04

Reproducibility note

The values are deliberately synthetic. They are internally consistent enough to support calculation and critique, but they must never be presented as experimental evidence.

2. Results and interpretation

The linear series has maxima near 30 and 210 degrees and minima approximately 90 degrees away. The circular series changes by less than five percent across analyzer angle. Offset subtraction increases the estimated linear visibility from 0.86 to 0.91, showing why baseline treatment matters.

Angle (deg)	Linear	Circular
0	0.78	0.50
30	1.00	0.51
60	0.76	0.49
90	0.26	0.50
120	0.05	0.52
150	0.27	0.48
180	0.79	0.50

Alternative explanations

A flat scan does not by itself prove perfect circular polarization. Rapid polarization drift, analyzer misalignment, detector saturation, or averaging over unresolved states can produce a similar pattern.

3. Discussion prompts

Use these prompts to test follow-up discussion in the Paper Guide workspace:

- Derive the visibility after subtracting the stated offset.
- Explain why a flat analyzer scan is necessary but not sufficient evidence of circular polarization.
- Propose one calibration and one control measurement.
- Compare the table with Malus' law and identify the largest residual.
- Rewrite the conclusion with a clear separation between observation, inference, and limitation.

Conclusion

The synthetic data reproduce the expected qualitative contrast between linear and circular states. A defensible interpretation still depends on detector linearity, analyzer calibration, repeatability, and explicit uncertainty reporting.